

Ben Parrack

(804) 221-0732 | benparrack@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Virginia Tech | Blacksburg, VA

Expected May 2028

Bachelor of Science in Computer Science

GPA: 3.5/4.0

Major in Data-Centric Computing, Minors in Math and Biomedical Engineering

- **Relevant Coursework:** Data Structures and Algorithms, Computer Organization, Data Analytics and Visualization, Applied Combinatorics, Discrete Mathematics, and Linear Algebra
- **Honors:** Dean's List (every semester)

TECHNICAL SKILLS

Languages: Java, Python, C, Bash, Ladder Logic

Tools and Technologies: Git, Linux, VMware Workstation, RSLogix, MATLAB, Arduino, matplotlib, SOLIDWORKS

EXPERIENCE

GrayMatter Systems | Ashland, VA | Automation Intern

May 2026 - August 2026

- Installed PLCs on-site, rewired them inside the panel and updated the installed Ladder Logic program
- Configured PLC network settings via Rockwell software, allowing communication with SCADA systems
- Designed and implemented a flexible inventory management system for GrayMatter's warehouse according to specified constraints (flexible, easily updatable, M365 compatible)

GenAI and AWS Bedrock Training and Jam | Blacksburg, VA

Sept 2025

- Completed AWS training course and Jam on GenAI services, including Amazon Bedrock, SageMaker, and various other applications (awarded certificate of completion for both)
- Gained valuable understanding of prompt engineering and responsible AI principles

PROJECTS

Interdisciplinary Project (IDPro@VT) - Hydroponics Design Team | Electronics Lead

Aug - Dec 2025

- Developed a hydroponics control system that integrated Arduino-based sensor monitoring and pump control to regulate nutrient delivery for optimized growth
- Configured instrumentation such as pH sensors and TDS probes to output human-readable values in ArduinoIDE
- Created project summary poster and presented with my team to 50+ people

VT Gym Occupancy Graph Program | Personal Project

Sept 2025

- Built a Python web scraper that pulled gym occupancy data every 5 minutes and visualized trends to help students optimize timing of gym visits
- Cached and accumulated data with pickle and graphically visualized results with matplotlib

LLM-Assisted Evaluation of Classroom Layouts and Movement Behaviors | Research

Jan - May 2026

- Reviewed minute-long segments of structured learning environment videos and compared LLM findings to the actual scenarios, finding key differences to base improvements for the model's behavior
- Analysis informed prompt refinement for classifying different behaviors and postures, with the end goal of improving student attentiveness

VOLUNTEER EXPERIENCE

CS² (Computer Science Community Service) | Blacksburg, VA | Executive Board Member

Aug 2025 - Present

- Led a program that taught 8 middle schoolers how to use Arduinos by guiding them through the process of building a programmable remote-controlled car over the course of a 6-week project (meeting weekly)
- Tutored 10 elementary school students weekly on how to create working programs in Scratch
- Designed curriculum and taught middle schoolers how to design objects in CAD and 3D print their designs